

BABU BANARASI DAS UNIVERSITY



SCHOOL OF ENGINEERING

Department of computer science & Engineering

PRIVACY AND SECURITY IN IOT LAB

(ITBC3751)

Session 2025-2026

PRACTICAL LAB FILE

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INDEX

S.NO.	Name of experiments	Page no.	Sign/Remark
1	Stenography using cmd	1-4	

OBJECTIVE

To hide text data inside an image file using the echo command in Command Prompt (CMD) as a basic form of steganography.

System Requirements:

Operating System: Windows (any version with Command Prompt)

Tools Required:

Command Prompt (built-in)

A JPG or PNG image file (e.g., photo.jpg)

Storage Location:

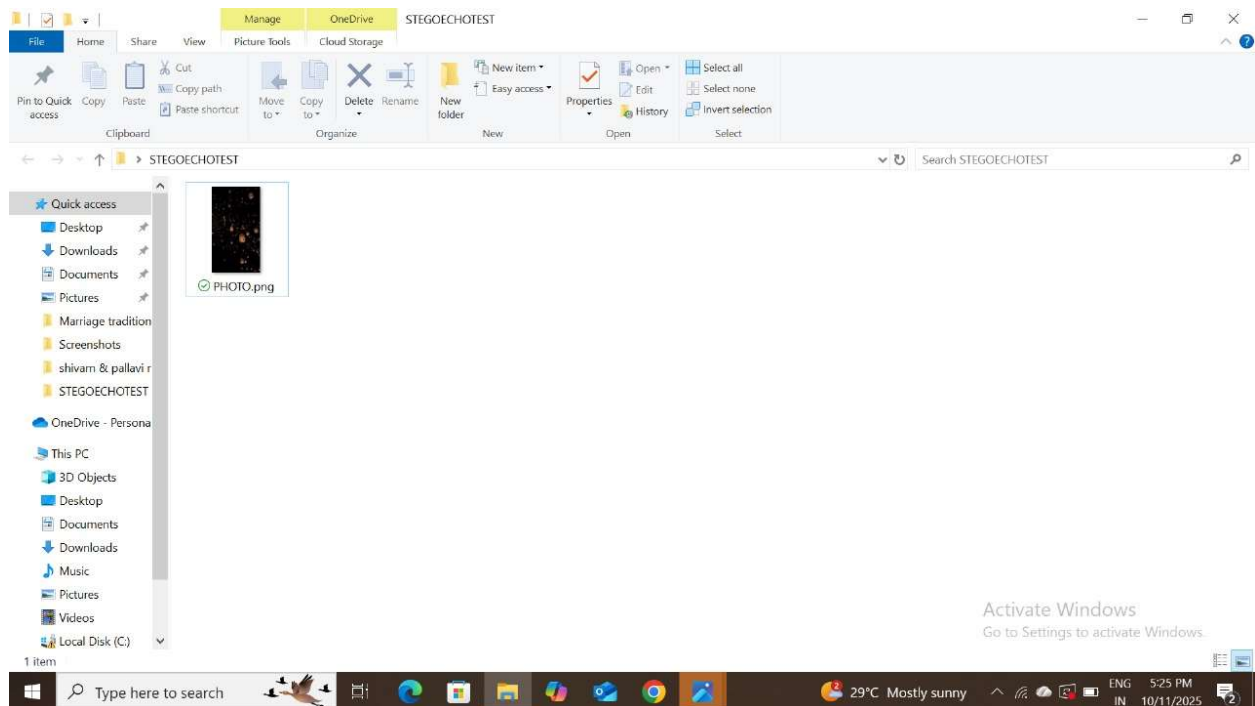
A working directory (e.g., C:\StegoEchoTest)

Procedure:

Step 1: Create and prepare the working folder

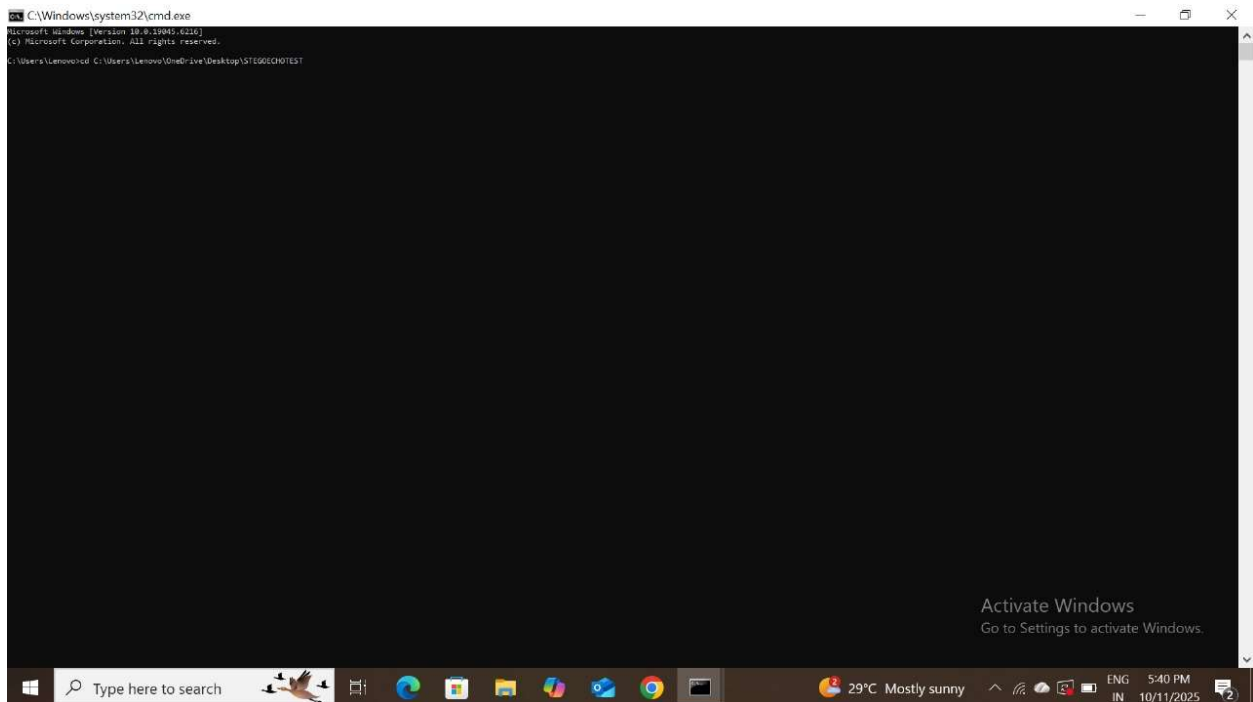
1. Create a folder named StegoEchoTest in C:\users\Dell\Desktop

2. Place the image file you want to use (e.g., photo.png) into this folder.



STEP 2-Open Command Prompt and navigate to the folder

1. Press Win + R, type cmd, and press Enter to open Command Prompt.
2. Navigate to your working directory:
3. cd C:\StegoEcho Test



STEP 3 -Hide a secret message using echo command

Run the following command to append a message to the image file:

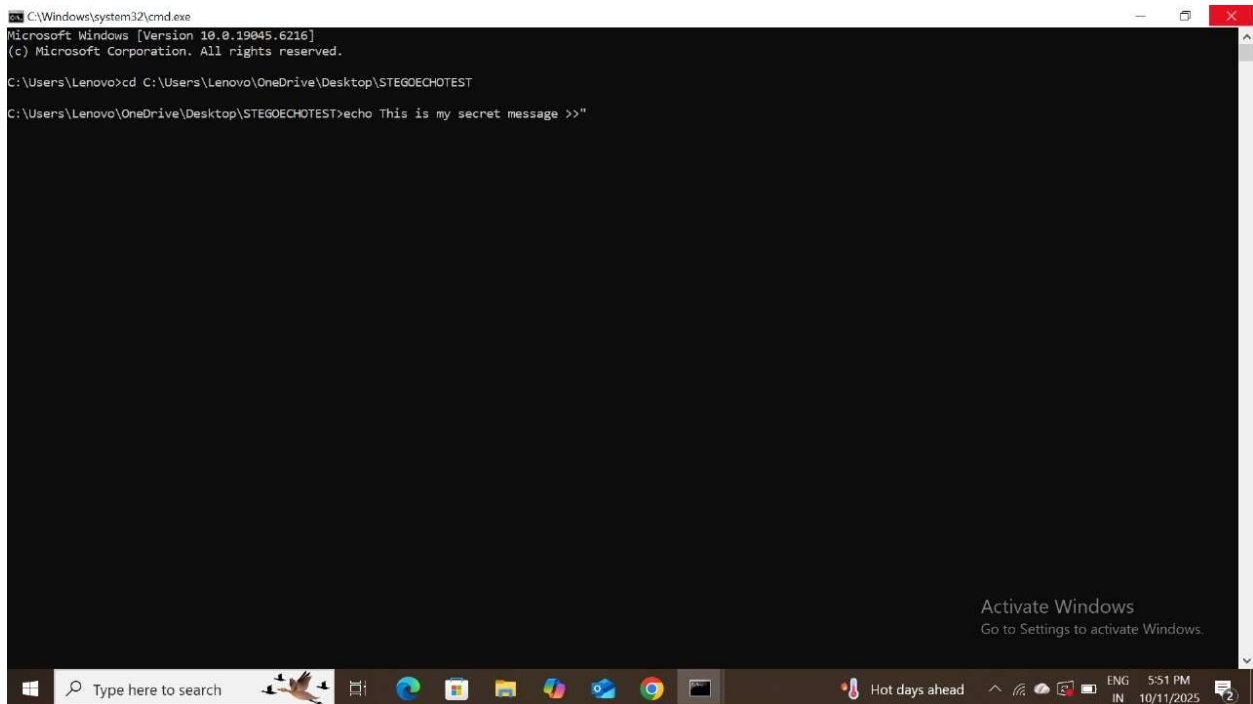
```
echo This is a secret message >> photo.jpg
```

What this does:

echo prints the message.

>> appends it to the end of photo.jpg in binary mode

.The image remain viewable in any any image viewer, but now contains hidden data.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>cd C:\Users\Lenovo\OneDrive\Desktop\STEGOECHOTEST
C:\Users\Lenovo\OneDrive\Desktop\STEGOECHOTEST>echo This is my secret message >>"
```

The screenshot shows a Windows Command Prompt window. The title bar indicates the path C:\Windows\system32\cmd.exe. The window content shows the following text: Microsoft Windows [Version 10.0.19045.6216] (c) Microsoft Corporation. All rights reserved. C:\Users\Lenovo>cd C:\Users\Lenovo\OneDrive\Desktop\STEGOECHOTEST C:\Users\Lenovo\OneDrive\Desktop\STEGOECHOTEST>echo This is my secret message >>" The Windows taskbar is visible at the bottom, showing the search bar, task view button, and several application icons. The system tray on the right shows the date and time as 5:51 PM on 10/11/2025, along with language and network indicators.

Step 4-Verify the hidden message

To check whether the message was added:

Option 1: Open with Notepad

```
notepad photo.jpg
```

Scroll down to the bottom to see the hidden message.

Option 2: Use more command

```
more < photo.jpg
```

This shows readable parts of the file including any appended text.

